Report on *Matrix Theory Reloaded: A BPS Road to Holography*

Blair, Lahnsteiner, Obers, Yan (IFT-UAM/CSIC-24-139; NORDITA 2024-033)

Scope. This report is based on the introduction, the holography and DLCQ-conjecture sections, the $T\bar{T}$ sections, the flow-equation summary and the Outlook of `paper.tex`. I did not work through the matrix-theory sections (2 and 3) in detail.

1. Principal achievement

The paper shows that the BPS decoupling limit that gives matrix theory on a Dp-brane (MpT) and the near-horizon limit that gives the bulk AdS geometry are the same operation, seen from two sides. Applying the MpT limit at asymptotic infinity of a Dp-brane background induces the near-horizon limit in the bulk. For $p = 3$ this reproduces the Maldacena limit, and the near-horizon $\mathrm{AdS}_5 \times S^5$ geometry is shown to asymptotically approach the flat non-Lorentzian M3T geometry. This gives a geometric and unified way to classify holographic duals.

The paper also derives p-brane generalisations of the $T\bar{T}$ flow. The general flow equation is

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \frac{1}{2\lambda^2}\Big\{\mathrm{tr}(\mathbb{1} - \lambda\tilde{T}) - (p-1)\big[\det(\mathbb{1} - \lambda\tilde{T})\big]^{\frac{1}{p-1}} - 2\Big\},$$

which gives the explicit $p = 2$ equation $\frac{1}{4}[\mathrm{tr}(\tilde{T}^2) - (\mathrm{tr}\,\tilde{T})^2] + \frac{\lambda}{2}\det\tilde{T}$. The solution for D free scalars is the Nambu–Goto-type Lagrangian $\mathcal{L} = \lambda^{-1}\big[1 - \sqrt{-\det(\eta + \lambda\,\partial X \partial X)}\big]$. I checked the $p = 2$ reduction by hand and the $p = 1$ trace form numerically: for the Nambu–Goto solution, $-\mathrm{tr}\,\tilde{T}/(2\lambda)$ equals $\frac{1}{2}\det T$ at a generic test point (two scalars). So the statement that the $p = 1$ case reduces to $T\bar{T}$ holds on this solution.

Finally, the paper conjectures a $\mathrm{DLCQ}^n/\mathrm{DLCQ}^m$ correspondence ($m > n$). In this scheme, $\mathrm{AdS}_5/\mathrm{CFT}_4$ is $\mathrm{DLCQ}^0/\mathrm{DLCQ}^1$, and the D1–D5 $\mathrm{AdS}_3/\mathrm{CFT}_2$ case is $\mathrm{DLCQ}^0/\mathrm{DLCQ}^2$.

2. Principal drawback

The headline claim, that the bulk geometry is generated *intrinsically* by a p-brane $T\bar{T}$ deformation, is not derived. The deformation parameter ω is promoted by hand to a function $\Omega(r)$. Its profile is then identified with the known Dp-brane near-horizon solution, $\Omega = (r/\ell)^{(7-p)/2}$ (Eq. (mptdtg)). The flow equations do not themselves determine this profile. The Outlook states that the authors did not verify that these geometries solve the MpT supergravity equations of motion and torsion constraints, so the intrinsic construction remains a proposal.

Two further limitations reduce the strength of the results:

- The flow equations are classical, bosonic and abelian. They are complete only for $p = 1, 2$. For $p = 3$ they are given to linear order in λ, and for $p > 3$ only the zeroth-order terms are given. Supersymmetric and non-abelian versions are left open.

- The DLCQ^n/DLCQ^m correspondence is a conjecture. Its support consists of matching geometries and classifying examples. I found no quantitative test, such as a spectrum or correlator comparison across the claimed duality. The case $n \geq 2$ is explicitly open.

3. Most worthwhile follow-up

Derive the bulk profile from the non-Lorentzian equations of motion instead of inserting it. Take the p-brane $T\bar{T}$-deformed ansatz (Eq. (mptsl022)) in the $p = 2$ (M2) case. This is the simplest flow equation, and the M2-brane non-relativistic M-theory analyses the paper cites for this regime (`Lambert:2024uue`, `Bergshoeff:2024nin`) provide the equations. Keep $\Omega(r)$ as an unknown function, substitute into the equations of motion and torsion constraints, and ask whether they force the power-law profile of the near-horizon solution, or exclude it.

- *Feasibility:* the task is a symbolic calculation in a computer-algebra system, using equations that already exist in the literature. It fits a few weeks for one researcher. The main risk is convention mismatch between the cited papers, which would need careful checking.
- *Impact:* this turns the paper's central conceptual claim, the answer to Question 3, into a checked statement. If the profile is forced, the intrinsic construction is supported in a concrete case. If it is obstructed, the result identifies which part of the proposal needs modification. Either outcome bears on the DLCQ classification and on non-Lorentzian holography more generally.

Alternatives considered: a full non-abelian or supersymmetric p-brane flow would be more general but much harder. A quantitative test of the DLCQ^n/DLCQ^m conjecture would need strong-coupling field-theory input and is likely out of reach in the near term.